Physical Chemistry Cume

Conversion factors: 1 atm = 101325 Pa; 1 bar = 100000 Pa

You may find the Clausius-Clapeyron equation useful:

$$\ln\frac{P_2}{P_1} = -\frac{\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

1) Argon has a normal melting point at 83.8K and a normal boiling point at 87.3K. It has a triple point at 0.7 atm and 83.8K and a critical temperature and pressure of 151 K and 48 atm.
   (a) In which phase is Argon under the following conditions: (i) 0.7 atm, 81 K; (ii) 1.4 atm, 86 K; (iii) 0.5 atm, 84 K; (iv) 48 atm, 150 K; (v) 1 atm, 87 K.
   (b) Which phase transitions occur when you do the following: (i) heat from 87 K to 89 K at 1atm; (ii) cool from 85K to 30 K at 0.69 atm, (iii) increase pressure from 0.5 atm to 1 atm at 82 K; (iv) decrease pressure from 3 atm to 0.2 atm at 83.8K.

2) In the nuclear industry, mined uranium ore (yellowcake, U_3O_8) is converted into uranium hexafluoride (hex, UF_6). UF_6 is used for uranium enrichment (by gaseous diffusion or in a gas centrifuge) because, unlike most other uranium compounds, UF_6 has a measurable vapor pressure for both the solid and liquid phases. The temperature dependence of the vapor pressure (in Pa) for the solid and liquid phases are given by:

$$\ln P_{solid} = 29.411 - 5893.5/T$$

$$\ln P_{liquid} = 22.254 - 3479.9/T$$

   (a) Calculate the temperature and pressure of the triple point of UF_6.
   (b) (i) In which physical state is UF_6 at room temperature and standard pressure (25°C and 1 bar)?
   (ii) Describe what happens as you heat up UF_6 from 25°C to 100 °C at standard (1 bar) pressure.

3) Carbon tetrachloride melts at 250 K. The vapor pressure of the liquid is 10539 Pa at 290K and 74518 Pa at 340 K. The vapor pressure of the solid is 270 Pa at 232K and 1092 Pa at 250 K.
   (a) Calculate $\Delta H_{vaporization}$, $\Delta H_{sublimation}$ and ΔH_{fusion}.
   (b) Calculate the normal boiling point (i.e. at 1 atm pressure).